Conduction, convection, radiation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify mechanisms of heat transfer and support each classification.

SUCCESS CRITERIA - I can...

1. Describe contact transfer, bulk fluid motion and electromagnetic transfer.
2. Use specific details or measurements from the situation.
3. Recognize that more than one mechanism can act.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A device transfers **15 J** in **3 s**, then performs the same transfer in **6 s**. **Find both average powers** and **explain the difference**.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $E_i + E_{in} = E_f + E_{out}$.
Account for energy across the system boundary. Conduction: matter interactions; convection: fluid motion; radiation: electromagnetic waves.

Worked reasoning: A metal rod transfers energy from its hot end toward its cool end by conduction. A circulating fluid can carry energy by convection; radiation can cross a vacuum.

Your turn: 1. **Draw a separate**, labeled **diagram** for each way energy is transferred.
2. **Show the direction of each** energy transfer.

Conduction	Convection	Radiation
Sketch:	Sketch:	Sketch:
_____	_____	_____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Classify: a metal spoon warming in soup; circulating water in a pot; sunlight warming your hand. **Give a reason** for each.

Situation	Mechanism	Evidence
Spoon in soup	____	____
Circulating water	____	____
Sunlight on skin	____	____

Marzano check - circle: ① ② ③

04 Apply the physics

A pot of water is heated from below. On the diagram, draw a circulation loop: warmer, less-dense water rising and cooler water sinking. **Explain how this bulk motion** differs from energy transfer along a solid metal spoon.



Heater below pot

Marzano check - circle: ① ② ③

Conduction, convection, radiation

antocciphysics

Continue down each column.

05 Explain your reasoning

An asteroid travels through nearly empty space. Which of these mechanisms can transfer thermal energy to it from sunlight? Why not ordinary air convection?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A rolling object loses **24 J** gravitational energy but gains only **16 J** translational kinetic energy. **Name two possible destinations of** the difference.
Show the energy changes and transfers; **label units**.

Initial | **Transfers** | **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

Explain how energy can travel from the Sun to Earth through space.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Design insulation for a hot cup using one change for each transfer mechanism.
2. **Explain each change**.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3